Unit 2 Assignment 3

Eigenvalues & Eigenvectors

Learning Objectives

By completing this assignment, your team will be able to:

- Understand eigenvalues and eigenvectors.
- Solve the characteristic equation manually.
- Compute eigenvectors manually and using NumPy.
- Interpret eigenvectors geometrically.
- Understand diagonalization.
- Perform eigen decomposition using NumPy.
- Explain the relationship between Eigenvalues and PCA.

Instructions

- Work collaboratively as a team.
- Show all manual calculations.
- Verify every numerical result using NumPy.
- Answer the AI Insight questions.
- Submit one notebook and one report per group.

Mission 1 – Discovering Eigenvectors

Consider the matrix

```
|2 1|
|1 2|
```

Suppose the following vectors are given:

```
v₁ = [1,1]

v₂ = [1,-1]
```

Tasks

1. Multiply the matrix by $\mathbf{v_1}$.
2. Multiply the matrix by $\mathbf{v_2}$.
3. Compare the direction of each vector before and after transformation.
4. Determine which vectors remain on the same direction.
5. Explain why these vectors are called Eigenvectors.

AI Insight

Why do most vectors change direction while only a few remain on the same line?

Mission 2 – Finding Eigenvalues

Using the same matrix

```
|2 1|
|1 2|
```

Tasks

1. Write the characteristic equation.
2. Compute

```
det(A − λI)=0
```

1. Solve the characteristic polynomial manually.
2. Find all eigenvalues.
3. Verify your answers using NumPy.

AI Insight

What does an eigenvalue represent during a matrix transformation?

Mission 3 – Finding Eigenvectors

Using the eigenvalues obtained previously,

Tasks

1. Compute the eigenvector for each eigenvalue manually.
2. Verify your answers using

```
np.linalg.eig()
```

1. Normalize each eigenvector (optional challenge).

AI Insight

Why does every eigenvalue have its own eigenvector?

Mission 4 – Geometric Interpretation

Assume a transformation produces the following results:

Eigenvalue	Interpretation
3	?
0.5	?

Eigenvalue	Interpretation
-2	?

Tasks

Explain what happens to the corresponding eigenvector when:

- λ = 3
- λ = 0.5
- λ = -2

AI Insight

How do eigenvalues affect scaling and direction?

Mission 5 – Matrix Diagonalization

For the matrix

```
|2 1|
|1 2|
```

Suppose

```
P
```

is the matrix of eigenvectors and

```
D
```

is the diagonal matrix of eigenvalues.

Tasks

1. Construct matrix **P**.

2. Construct matrix **D**.
3. Verify using NumPy that

$$A = P D P^{-1}$$

1. Explain why diagonal matrices are easier to work with.

AI Insight

Why is diagonalization useful in Machine Learning?

Mission 6 – Eigenvalues in AI

A data scientist wants to reduce the number of features before training a Machine Learning model.

Tasks

Answer the following questions:

1. Why is PCA used?
2. Why are eigenvalues important in PCA?
3. Why do we sort eigenvalues from largest to smallest?
4. Which eigenvectors should be selected?
5. What may happen if we remove the eigenvector with the largest eigenvalue?

AI Insight

Explain how Eigenvalues help reduce data dimensions while preserving useful information.

Submit:

✅ Jupyter Notebook

✅ PDF Report